

SCRUM MEETING REPORT

Project Name: Flood Monitoring System (FMS)

Sprint Number: 6

Date: April 11, 2026

Time: 10:30 AST (UTC-3) Location: Halifax, Nova Scotia

Participant • Benjamin (NSCC), Keaton (NSCC), Kyran (NSCC), Abdullah (SpaceXAI), Karthik (SpaceXAI), Mansi (SpaceXAI)

Meeting Objectives

1. Update the team on project status
2. Presented final deliverable
3. Answered questions on pre-processing and model selection
4. Confirmed with industry partner that requirements were met.

Meeting Overview

The meeting focused on presenting the team's remote sensing project that aimed to detect flooding in satellite imagery using binary classification in central Bangladesh's Dhaka region. Kyran explained the data pipeline using Sentinel-1 and Sentinel-2 data with specific bands, while Keaton detailed the model architecture and performance comparison of six different models including logistic regression, random forest, XGBoost, SVM, CNN, and ResNet 18, with XGBoost achieving the best AUC score of 0.99. The team discussed feature leakage concerns and the potential benefits of using CNN models. Karthik expressed satisfaction with the methodology work completed and mentioned plans to begin drafting the final research paper using LaTeX formatting, with M offering to assist with the LaTeX formatting process.

Key Discussion Points

Data Pipeline & Preprocessing

- **Status:** Kyran has **completed** the automated pipeline for downloading data to cloud storage. Automated preprocessing is done.

Model Selection

- **Status:** Keaton has **completed** the model development. XGBoost and CNN performed the best. We have chosen to stick with CNN for the final paper.

Next Steps: Begin drafting final research paper using LaTeX formatting.

1. Action Items

<i>Table 1. Action Items</i>	Assigned To	Due Date	Status
Action Item			
Finalize written system scope description	Benjamin	02/23/26	X Complete
Complete DFD Level 0	Benjamin	02/23/26	X Complete
Complete DFD Level 1	Benjamin	02/24/26	X Complete
Draft ERD structure	Benjamin	02/25/26	X Complete
Prepare initial SQL database script	Benjamin	02/26/26	X Complete
Automatically download files	Kyran	03/22/26	X Complete
Transform the data via preprocessing	Kyran	03/28/26	X Complete
Build Models and Test Performance	Keaton	04/01/26	X Complete
Visualize the Findings	Nathan	04/07/26	X Complete
Present to NSCC	Everyone	04/07/26	X Complete
Present to SpaceXAI	Everyone	04/03/26	X Complete

Project has been completed